

DEF (INTEGRALE): SIA $[a, b]$ INT. CHIUSO E LIMITATO, $f: [a, b] \rightarrow \mathbb{R}$ FUNZIONE LIMITATA.

SI DICE CHE f È **INTEGRABILE** SU $[a, b]$ CON INTEGRALE $I \in \mathbb{R}$ SE:

$\forall \varepsilon > 0 \exists \tilde{m} \in \mathbb{N}$ t.c. $\forall n \geq \tilde{m}$ ALLORA:

$$|S(f, \xi, n) - I| < \varepsilon$$

E TALE $\tilde{m}$ NON DIPENDE DA ε .

IN TAL CASO SI PONE: $\int_a^b f dx = I$

TEO (COND. SUFF. INTEGRABILITÀ): $f: [a, b] \rightarrow \mathbb{R}$.

1) SE f È MONOTONA $\rightarrow f$ INTEGRABILE

2) SE f È CONTINUA $\Rightarrow f$ INTEGRABILE

3) SE f È MONOTONA O CONTINUA A TRATTI $\Rightarrow f$ È INTEGRABILE

DEF (INTEGRALE SU INTERVALLI ORIENTATI): SE $f: [a, b] \rightarrow \mathbb{R}$ È INTEGRABILE, PONIAMO:

$$\int_a^b f dx = - \int_b^a f dx$$

TEO (PROP. DELL'INTEGRALE): $f, g: [a, b] \rightarrow \mathbb{R}$ INTEGRABILI.

1) SE $\alpha, \beta \in \mathbb{R}$, ALLORA: $\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx$

2) SE $c \in [a, b]$: $\int_a^b f dx = \int_a^c f dx + \int_c^b f dx$

3) $f \leq g$ ALLORA: $\int_a^b f dx \leq \int_a^b g dx$

4) $|\int_a^b f dx| \leq \int_a^b |f| dx$

DEF (PRIMITIVA): $I \subseteq \mathbb{R}$, $f: I \rightarrow \mathbb{R}$, $F: I \rightarrow \mathbb{R}$ DERIVABILE.

F SI DICE **PRIMITIVA** DI f SE: $F'(x) = f(x) \quad \forall x \in I$

TEO (FONDAMENTALE DEL CALCOLO INTEGRALE): $f: [a, b] \rightarrow \mathbb{R}$ CONTINUA E F UNA PRIMITIVA DI f . ALLORA $\int_a^b f dx = F(b) - F(a) \quad (= |F(x)|_a^b)$

TEO: f, g DERIVABILE SU (a, b) CONT. SU $[a, b]$. ALLORA:

$$\int_a^b f' g dx = fg - \int f g'$$

$$\bullet \int \cos(x) dx = \sin(x) + C \quad \bullet \int \sin(x) dx = -\cos(x) + C$$

$$\bullet \int e^x = e^x + C \quad \bullet \int \frac{1}{x^2+1} dx = \arctan(x) + C$$

$$\bullet \int \sinh(x) dx = \cosh(x) + C \quad \bullet \int \cosh(x) dx = \sinh(x) + C$$

$$\bullet \int \left(\frac{2}{x} - x^{33} + e^x \right) dx = 2 \log|x| - \frac{x^{34}}{34} + e^x + C$$

$$\bullet \int e^{16x} dx = \frac{1}{16} \int 16 e^{16x} dx = \frac{1}{16} e^{16x} + C$$

$$\bullet \int e^{\frac{x}{2}} dx = 2 \int \frac{1}{2} e^{\frac{x}{2}} dx = 2 e^{\frac{x}{2}} + C$$

$$\bullet \int \frac{x}{4x^2+1} dx = \frac{1}{8} \int \frac{8x}{4x^2+1} dx = \frac{1}{8} \log|4x^2+1| + C$$

$$\bullet \int \sin(x) \cdot \cos(x) dx = \frac{\sin^2(x)}{2} + C$$

$$\bullet \int \frac{1}{x^2+2} dx = \frac{1}{2} \int \frac{1}{\left(\frac{x}{\sqrt{2}}\right)^2+1} dx = \frac{\sqrt{2}}{2} \int \frac{\frac{1}{\sqrt{2}}}{\left(\frac{x}{\sqrt{2}}\right)^2+1} dx = \frac{\sqrt{2}}{2} \arctan\left(\frac{x}{\sqrt{2}}\right) + C$$

$$\bullet \int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx = - \int \frac{1}{\cos(x)} \cdot \underbrace{(-\sin(x))}_{\text{DERIVATA DI } \cos(x)} dx = -\log|\cos(x)| + C$$

$$\bullet \int \sinh(2x) dx = \frac{1}{2} \int \sinh(2x) \cdot 2 dx = \frac{1}{2} \cosh(2x) + C$$

$$\bullet \int \sqrt{6x-2} dx = \int (6x-2)^{\frac{1}{2}} dx = \frac{1}{6} \int 6 (6x-2)^{\frac{1}{2}} dx = \frac{1}{6} \cdot \frac{2}{\frac{3}{2}} (6x-2)^{\frac{3}{2}} + C$$

$$= \frac{1}{9} \sqrt{(6x-2)^3} + C$$

$$\bullet \int_0^{\pi} x \sin(x) dx = \left(-x \cos(x) \right) \Big|_0^{\pi} - \int_0^{\pi} -\cos(x) dx$$

$$= (-\pi \cdot (-1)) + \int_0^{\pi} \cos(x) dx =$$

$$= \pi + \sin(x) \Big|_0^{\pi} = \pi + (0 - 0) = \pi$$

$$\begin{array}{ccc} \int & & D \\ 1 & \longleftarrow & x \\ & \searrow & \downarrow \\ \sin(x) & \longrightarrow & -\cos(x) \end{array}$$

$$\begin{aligned} \bullet \int \log(x) dx &= x \log(x) - \int \frac{1}{x} \cdot x dx = \\ &= x \log(x) - x + C = x(\log(x) - 1) + C \end{aligned}$$

$$\begin{array}{ccc} \int & & \Delta \\ \frac{1}{x} & \xleftarrow{\quad} & \log(x) \\ & \xrightarrow{\quad} & \downarrow \\ 1 & \xrightarrow{\quad} & x \end{array}$$

$$\begin{aligned} \bullet \int \arctan(x) dx &= x \arctan(x) - \int \frac{x}{x^2+1} dx = \\ &= x \arctan(x) - \frac{1}{2} \int \frac{2x}{x^2+1} dx = \\ &= x \arctan(x) - \frac{1}{2} \log|x^2+1| + C \end{aligned}$$

$$\begin{array}{ccc} \int & & \Delta \\ \frac{1}{x^2+1} & \xleftarrow{\quad} & \arctan \\ & \xrightarrow{\quad} & \downarrow \\ 1 & \xrightarrow{\quad} & x \end{array}$$

$$\begin{aligned} \bullet \int x^2 e^{2x} dx &= \frac{1}{2} x^2 e^{2x} - \int \frac{1}{2} 2x e^{2x} dx = \\ &= \frac{1}{2} x^2 e^{2x} - \int x e^{2x} dx = \\ &= \frac{1}{2} x^2 e^{2x} - \left(\frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx \right) \\ &= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C \end{aligned}$$

$$\begin{array}{ccc} \int & & \Delta \\ 2x & \xleftarrow{\quad} & x^2 \\ e^{2x} & \xrightarrow{\quad} & \downarrow \\ 1 & \xleftarrow{\quad} & x \\ e^{2x} & \xrightarrow{\quad} & \frac{1}{2} e^{2x} \end{array}$$

$$\begin{aligned} \bullet \int_0^4 \sin(\sqrt{x}) dx &\Rightarrow \begin{array}{l} t = \sqrt{x} \\ x = t^2 \\ dx = 2t dt \\ x=4 \Rightarrow t = \sqrt{4} = 2 \\ x=0 \Rightarrow t = 0 \end{array} \Rightarrow \int_0^2 \sin(t) 2t dt = \begin{array}{ccc} \int & & \Delta \\ 2 & \xleftarrow{\quad} & 2t \\ \sin(t) & \xrightarrow{\quad} & -\cos(t) \end{array} \end{aligned}$$

$$= -2t \cos(t) \Big|_0^2 + \int_0^2 2 \cos(t) dt = (-4 \cos(2) - 0) + 2(\sin(t)) \Big|_0^2 =$$

$$= -4 \cos(2) + 2(\sin(2) - 0) = 2(-2 \cos(2) + \sin(2))$$

$$\begin{aligned} \bullet \int_0^1 e^{16x} dx &\Rightarrow \begin{array}{l} t = 16x \\ x = \frac{1}{16} t \\ dx = \frac{1}{16} dt \\ x=0 \Rightarrow t=0 \\ x=1 \Rightarrow t=16 \end{array} \Rightarrow \int_0^{16} e^t \frac{1}{16} dt = \frac{1}{16} \int_0^{16} e^t dt = \\ &= \frac{1}{16} e^t \Big|_0^{16} = \frac{1}{16} (e^{16} - 1) \end{aligned}$$

$$\begin{aligned} \bullet \int \frac{e^{2x}}{\sqrt{e^x+1}} dx &\Rightarrow \begin{array}{l} t = \sqrt{e^x+1} \\ x = \log(t^2-1) \\ dx = \frac{1}{t^2-1} 2t dt \\ e^x = t^2-1 \\ e^{2x} = (t^2-1)^2 \end{array} \rightarrow \int \frac{(t^2-1)^2}{t^2-1} \frac{1}{t^2-1} 2t dt = 2 \int (t^2-1) dt = \\ &= \frac{2}{3} t^3 - 2t + C = \frac{2}{3} (e^x+1)^{\frac{3}{2}} - 2(e^x+1)^{\frac{1}{2}} + C \end{aligned}$$

$$\cdot \int_0^2 \frac{\sqrt[6]{x}}{\sqrt[3]{x}-2} dx \Rightarrow \begin{array}{l} \sqrt[6]{x} = t \\ x = t^6 \\ dx = 6t^5 dt \\ x=0 \Rightarrow t=0 \\ x=2 \Rightarrow t=\sqrt[6]{2} \\ \sqrt[3]{x} = t^2 \end{array} \Rightarrow \int_0^{\sqrt[6]{2}} \frac{t}{t^2-2} 6t^5 dt =$$

$$= 6 \int_0^{\sqrt[6]{2}} \frac{t^6}{t^2-2} dt *$$

DIVISIONE POLINOMI:

$$\begin{array}{r|l} t^6 & t^2-2 \\ -t^6+2t^4 & \\ \hline / +2t^4 & \\ -2t^4+4t^2 & \\ \hline / 4t^2 & \\ -4t^2+8 & \\ \hline / 8 & \end{array} \quad \begin{array}{l} t^2-2 \\ t^4+2t^2+4 \end{array} \Rightarrow \frac{t^6}{t^2-2} = t^4+2t^2+4 + \frac{8}{t^2-2}$$

$$* 6 \int_0^{\sqrt[6]{2}} (t^4+2t^2+4) dt + 8 \cdot 6 \int_0^{\sqrt[6]{2}} \frac{1}{t^2-2} =$$

$$= 6 \left(\frac{t^5}{5} + \frac{2}{3} t^3 + 4t \right) \Big|_0^{\sqrt[6]{2}} + 48 \int_0^{\sqrt[6]{2}} \frac{1}{t^2-2} =$$

$$= 6 \left(\frac{2^{\frac{5}{6}}}{5} + \frac{2}{3} 2^{\frac{1}{2}} + 4 \cdot 2^{\frac{1}{6}} \right) + 48 \int_0^{\sqrt[6]{2}} \frac{1}{t^2-2} *$$

$$* \frac{1}{t^2-1} = \frac{A}{t+\sqrt{2}} + \frac{B}{t-\sqrt{2}} = \frac{At+Bt-A\sqrt{2}+B\sqrt{2}}{(t+\sqrt{2})(t-\sqrt{2})}$$

$$\begin{cases} A+B=0 \\ -\sqrt{2}A+\sqrt{2}B=1 \end{cases} \Rightarrow \begin{cases} A=-B \\ \sqrt{2}B+\sqrt{2}B=1 \Rightarrow \sqrt{2}(2B)=1 \end{cases} \quad \begin{array}{l} A = -\frac{1}{2\sqrt{2}} \\ B = \frac{1}{2\sqrt{2}} \end{array}$$

$$* 48 \left(-\frac{1}{2\sqrt{2}} \right) \int_0^{\sqrt[6]{2}} \frac{1}{t+\sqrt{2}} + 48 \left(\frac{1}{\sqrt{2}} \right) \int_0^{\sqrt[6]{2}} \frac{1}{t-\sqrt{2}} =$$

$$= -\frac{48}{2\sqrt{2}} \left(\log |t+\sqrt{2}| \right) \Big|_0^{\sqrt[6]{2}} + \frac{48}{2\sqrt{2}} \left(\log |t-\sqrt{2}| \right) \Big|_0^{\sqrt[6]{2}}$$

$$* = 6 \left(\frac{2^{\frac{5}{6}}}{5} + \frac{2}{3} 2^{\frac{1}{2}} + 4 \cdot 2^{\frac{1}{6}} \right) - \frac{48}{2\sqrt{2}} \left(\log |t+\sqrt{2}| \right) \Big|_0^{\sqrt[6]{2}} + \frac{48}{2\sqrt{2}} \left(\log |t-\sqrt{2}| \right) \Big|_0^{\sqrt[6]{2}}$$

$$\int \frac{x+1}{x^3+x^2-2x} dx = \int \frac{x+1}{x(x^2+x-2)} dx = \int \frac{x+1}{x(x+2)(x-1)} dx = (*)$$

$$\frac{x+1}{x(x+2)(x-1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+2)} = \frac{A(x+2)(x-1) + Bx(x+2) + Cx(x-1)}{x(x+2)(x-1)} =$$

$$= \frac{A(x^2+x-2) + B(x^2+2x) + C(x^2-x)}{x(x+2)(x-1)} = \frac{x^2(A+B+C) + x(A+2B-C) - 2A}{x(x+2)(x-1)}$$

$$\begin{cases} A+B+C=0 \\ A+2B-C=1 \\ -2A=1 \end{cases} \Rightarrow \begin{cases} A=-\frac{1}{2} \\ -\frac{1}{2}+2B-C=1 \\ -\frac{1}{2}+B+C=0 \end{cases} \Rightarrow \begin{cases} -\frac{1}{2}+2B-C=1 \\ -1+3B=1 \Rightarrow B=\frac{2}{3} \\ -\frac{1}{2}+\frac{2}{3}=-C \\ \frac{-3+4}{6}=\frac{1}{6}=-C \\ C=-\frac{1}{6} \end{cases}$$

$$(*) -\frac{1}{2} \int \frac{1}{x} dx + \frac{2}{3} \int \frac{1}{(x-1)} dx - \frac{1}{6} \int \frac{1}{(x+2)} dx =$$

$$= -\frac{1}{2} \log|x| + \frac{2}{3} \log|x-1| - \frac{1}{6} \log|x+2| + C$$

$$\bullet \int \frac{1}{(x+1)(x-3)^2} dx = \frac{A}{x+1} + \frac{B}{(x-3)} + \frac{C}{(x-3)^2} =$$

$$\frac{A(x-3)^2 + B(x+1)(x-3) + C(x+1)}{(x+1)(x-3)^3} = \frac{A(x^2-6x+9) + B(x^2-2x-3) + C(x+1)}{(x+1)(x-3)^3}$$

$$= \frac{x^2(A+B) + x(-6A-2B+C) + 9A-3B+C}{(x+1)(x-3)^3}$$

$$\begin{cases} A+B=0 \\ -6A-2B+C=0 \\ 9A-3B+C=1 \end{cases} \Rightarrow \begin{cases} A=-B \\ 6B-2B+C=0 \\ -9B-3B+C=1 \end{cases} = (+3) \cdot \begin{cases} A=-B \\ 4B+C=0 \\ -12B+C=1 \end{cases} \Rightarrow \begin{cases} C=\frac{1}{4} \\ B=-\frac{1}{16} \\ A=\frac{1}{16} \end{cases}$$

$$= \int \frac{1}{(x+1)(x-3)^2} dx = \frac{1}{16} \int \frac{1}{x+1} dx - \frac{1}{16} \int \frac{1}{x-3} dx + \frac{1}{4} \int \frac{1}{(x-3)^2} dx$$

$$= \frac{1}{16} \log|x+1| - \frac{1}{16} \log|x-3| - \frac{1}{4} \left(\frac{1}{x-3} \right) + C$$

$$\bullet \int \frac{x^2}{x^2+4} dx = \int \left(1 - \frac{4}{x^2+4}\right) dx = \left. \frac{x^2}{1} - \frac{-x^2-4}{-4} \right|_{x^2+4}$$

$$= \int 1 dx - 4 \int \frac{1}{x^2+4} dx = \int 1 dx - \frac{4}{4} \int \frac{1}{\left(\frac{x}{2}\right)^2+1} d\left(\frac{x}{2}\right) =$$

$$= \int 1 dx - 2 \int \frac{\frac{1}{2}}{\left(\frac{x}{2}\right)^2+1} dx = x - 2 \arctan\left(\frac{x}{2}\right) + C$$

$$\bullet \int x^2 \log x dx = \frac{1}{3} x^3 \log(x) - \int \frac{1}{x} \cdot \frac{x^3}{3} dx$$

$\begin{matrix} D & \int \\ \frac{1}{x} & \leftarrow \log(x) \\ x^2 & \rightarrow \frac{x^3}{3} \end{matrix}$

$$= \frac{1}{3} x^3 \log(x) - \frac{1}{3} \int x^2 dx =$$

$$= \frac{1}{3} x^3 \log(x) - \frac{x^3}{9} + C$$

$$\bullet \int \frac{x+1}{x^3+x^2-2x} dx = \int \frac{x+1}{x(x^2+x-2)} dx = \int \frac{x+1}{x(x+2)(x-1)} dx = (*)$$

$$\frac{x+1}{x(x+2)(x-1)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-1} = \frac{A(x^2+x-2) + B(x^2-x) + C(x^2+2x)}{x(x+2)(x-1)} =$$

$$= \frac{x^2(A+B+C) + x(A-B+2C) - 2A}{x(x+2)(x-1)}$$

$$\begin{cases} A+B+C=0 \\ A-B+2C=1 \\ -2A=1 \end{cases} \quad \begin{cases} A=-\frac{1}{2} \\ -\frac{1}{2}+B+C=1 \\ -\frac{1}{2}-B+2C=0 \end{cases}$$

$$\begin{matrix} -1 & / & +3C=1 \end{matrix}$$

$$\begin{cases} A=-\frac{1}{2} \\ C=\frac{2}{3} \\ B=\frac{1}{2}-\frac{2}{3}=\frac{3-4}{6}=-\frac{1}{6} \end{cases}$$

$$* -\frac{1}{2} \int \frac{1}{x} dx - \frac{1}{6} \int \frac{1}{x+2} + \frac{2}{3} \int \frac{1}{x-1} = -\frac{1}{2} \log|x| - \frac{1}{6} \log|x+2| + \frac{2}{3} \log|x-1| + C$$

$$\bullet \int (2x-1) \sin(x) dx =$$

$$= (1-2x) \cos(x) + 2 \int \cos(x) dx =$$

$$\begin{matrix} D & \int \\ 2 & \leftarrow (2x-1) \\ \sin(x) & \rightarrow -\cos(x) \end{matrix}$$

$$= (1-2x) \cos(x) + 2 \sin(x) + C = \cos(x) - 2x \cos(x) + 2 \sin(x) + C$$